[2024-2025 春夏] 离散数学理论基础 Quiz3

悔教 @cc98

2025 年 7 月 1 日

1. (5% for each blank)

   (1) List all ordered pairs in the partial ordering with the accompanying Hasse diagram (see Fig.1).

   _____

   (2) How many spanning trees does $K_{2,n}$ have? _____

   (3) How many edges does Q_n have? _____

   (4) There are _____ non-isomorphic simple directed graphs with 3 vertices and 2 edges. There are _____ non-isomorphic rooted trees with 4 vertices.

   (5) In $K_{2,3}$, find the number of paths of length 3 between every pair of vertices. _____

   (6) Find the smallest equivalence relation on $\{1,2,3,4,5\}$ that contains $(1,2)$, $(1,4)$ and $(3,5)$.

   _____

   (7) List the relations on the set $\{a,b\}$ that are reflexive and symmetric. _____

   (8) Use Huffman coding to encode these symbols with given frequencies: a: 0.14, b: 0.23, c: 0.27, d: 0.18, e: 0.08, f: 0.10. What is the average number of bits required to encode a character?

   _____

2. (5% and 8%)

   (1) (5%) List all positive integers n such that Q_n has an Euler circuit. Explain why.

   (2) (8%) List all positive integers n such that Q_n has a Hamilton circuit. Explain why.

3. (8%) Find the transitive closure of R if the adjacency matrix M_R is

$$M_R = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

4. (8%) Calculate the maximum flow and minimum cut of the following network flow (see Fig.2).

5. (8%) Let G be a simple graph and $\chi(G)$ its chromatic number. Let k be the maximum degree among the vertices of G. Prove that

$$\chi(G) \leq k+1.$$

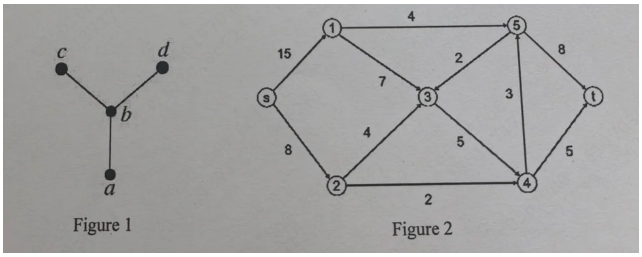


图 1: Fig 1 and Fig 2

6. (8%) A tournament is a simple directed graph such that for any two distinct vertices u and v, exactly one of (u, v) and (v, u) is an edge in the graph. Prove that, in a tournament, for any vertex S with maximum out-degree and for any other vertex T $(T \neq S)$, there exists a directed path from S to T.

7. (10%) The thickness of a simple graph G is defined as the minimum number of planar subgraphs whose union is G. Find the thickness of the graph $K_{4,5}$ and provide a proof for your answer.